Tec-C: 4-Person Engineering Roadmap

Decoupled Full-Stack Engineering Architecture & Daily Workstreams (Oct 16 – Oct 20)

Target Event: Shenzhen FinTechathon 2026 (International Track)	**Topic Track:** Topic F: Agent Wallet (Autonomous On-Chain Wallet)
Co-Hosts: Tsinghua University PBCSF & WeBank	**Submission Deadline:** October 20, 2026 (23:59 Beijing Time)
Team Structure: 100% Technical Engineering Roster (4 Leads)	**Team Compliance:** ≥ 60% non-Chinese members verified

1. Executive Sprint Strategy: Dual-Lead AI Architecture

To achieve maximum technical depth for the judging panel (Tsinghua PBCSF and WeBank AI), **the AI layer is decoupled across two dedicated engineers** rather than relying on a non-technical pitch lead. Member 1 builds the on-chain smart contract vault, Member 2 builds the cognitive planning and strategy engine, Member 3 builds the user dashboard and live attack sandbox, and Member 4 builds the PegGuard Layer 4 NLP sentinel and omni-market macro oracles. Pitch deck production, demo video recording, and final lodgement are shared across the team.

2. The 4-Person Technical Ownership Matrix

Role & Title	Primary Technical Domain	Core Deliverable Outputs	Target Judges Satisfied
Member 1: **Smart Contracts Lead** *The Vault Architect*	Solidity 0.8.24, Foundry, ERC-7579 Minimal Account, OpenZeppelin, Anvil fork.	• TecAccount.sol (modular vault) • TetherGuardHook.sol (immutable) • SessionKeyValidator.sol • TecPaymaster.sol (anti-griefing) • 100% Foundry Unit Tests	**Henry Ma & Kaixiang Zhang** (WeBank Account Abstraction & Security Standards)
Member 2: **Cognitive AI & Strategy Lead** *The Brain & Planner*	Python 3.11, FastAPI, LangGraph, Pydantic v2, Web3.py, Local Privacy Sieve.	• LangGraph Intent Parser • Dual-Axis Risk Profiler • Proactive Strategy Engine ('Good, Better, Best') • Structured UserOp Compiler • EIP-712 WeIdentity Signer	**Prof. Dacheng Xiu & Henry Ma** (Proactive Agency, Task Completion & Wealth UX)
Member 3: **Frontend & UX Lead** *The Radar Architect*	Next.js 14 (App Router), Tailwind CSS, Lucide Icons, Wagmi, Viem, WebAuthn Passkeys.	• Passkey Biometric Login (FaceID) • The Tether Radius Progress Gauge • Proactive Strategy Comparison Cards • Breaking the Tether Modal • **The Live Attack Sandbox**	**Evaluation Rubric (30% UX)** (Zero seed-phrase consumer banking experience)
Member 4: **Financial NLP & Oracles Lead** *The Sentinel & Macro Lead*	Hugging Face Transformers, ProsusAI/finbert, spaCy/NER, Shibor/Macro APIs, Anvil fork.	• PegGuard Layer 4 NLP Sentinel • FinBERT Adverse News Veto (≥60%) • Omni-Market Oracle (Shibor + Gold) • Pre-Flight Dry-Run Simulator • Shared Pitch & Video Co-Lead	**Prof. Qiang Yang, Dr. Lixin Fan, & Prof. Jianhua Zhang** (Trustworthy AI & PBOC Macro Standards)

3. Member 1 Workstream: Smart Contracts & On-Chain Security

Member 1 builds the on-chain vault that guarantees user funds cannot be drained even if the AI backend is compromised:

Day & Timeline	Engineering Tasks & Implementation Scope	Verification Gate & Deliverable
Day 1 (Oct 16) *Contracts Setup*	Initialize Foundry workspace. Implement TecAccount.sol (ERC-7579 minimal account). Implement TetherGuardHook.sol enforcing The Tether Radius ($500/day default limit) and asset tier whitelist at EVM bytecode level.	Compiles cleanly with forge build. Zero compilation warnings.
Day 2 (Oct 17) *Validator & Paymaster*	Implement SessionKeyValidator.sol verifying compact 32-byte EIP-712 credential commitments. Implement TecPaymaster.sol with 0.005 ETH session gas quota. Launch local Anvil testnet fork with live liquidity.	Contracts deployed to local Anvil testnet; addresses logged.
Day 3 (Oct 18) *Invariant Testing*	Write comprehensive Foundry unit/fuzz tests verifying: (1) Legitimate swap executes under limit; (2) Breaching Tether Radius reverts with TetherRadiusExceeded(); (3) Unauthorized address reverts with UnauthorizedTarget(); (4) Gas griefing halts at 0.005 ETH.	**100% Green Tests:** forge test -v passing all tests. Export ABIs to /shared/contracts.json.
Day 4 (Oct 19) *Gas Profiling*	Run automated gas profiling reports (forge test --gas-report) proving ~3,800 gas verification. Provide transaction hashes and contract addresses to team for video recording.	Gas benchmark report generated; contract addresses shared.
Day 5 (Oct 20) *Code Freeze*	Add comprehensive NatSpec docstrings; clean commit history; tag stable release on GitHub repository.	Smart contract layer frozen and public on GitHub.

4. Member 2 Workstream: Cognitive AI & Proactive Strategy Engine

Member 2 builds the intent state machine, local privacy sieve, and the proactive multi-option strategy generator:

Day & Timeline	Engineering Tasks & Implementation Scope	Verification Gate & Deliverable
Day 1 (Oct 16) *FastAPI Scaffolding*	Scaffold /agent repository with Python 3.11, FastAPI, and LangGraph. Implement natural-language intent parser translating commands (*'Stake my 10 ETH'*) into typed Pydantic models. Implement local PII masking (redacting balances and user identity).	POST /api/mandate/parse returns structured Pydantic JSON.
Day 2 (Oct 17) *Dual-Axis Risk Engine*	Build the Dual-Axis Risk Profiler calibrating Axis 1 (Asset Universe: Conservative vs Balanced vs Aggressive) and Axis 2 (The Tether Radius: $250 vs $1,000 vs $2,500). Integrate mandate validator ensuring operations respect asset tiers.	Risk parameters successfully configure session key scopes.
Day 3 (Oct 18) *Proactive Strategy*	Build the 'Good, Better, Best' recommendation engine (evaluating Lido staking @ 3.1% vs Curve stETH/ETH LP @ 6.85%). Build UserOp encoder signing EIP-712 WeIdentity credential digests. Connect to Member 1's Anvil testbed.	UserOp formatted with valid EIP-712 credential signatures.
Day 4 (Oct 19) *Backend Integration*	Finalize core endpoints for Member 3's UI: /api/mandate/parse, /api/mandate/propose, and /api/mandate/execute. Connect with Member 4's NLP risk veto feed.	End-to-end intent to execution flow operational.
Day 5 (Oct 20) *Docs & Freeze*	Package Python dependencies in requirements.txt; generate Swagger OpenAPI documentation (/docs).	Backend API fully documented and reproducible.

5. Member 3 Workstream: Frontend UI, Biometrics & Attack Sandbox

Member 3 delivers the visual dashboard, Passkey WebAuthn login, and the interactive Attack Sandbox for judges:

Day & Timeline	Engineering Tasks & Implementation Scope	Verification Gate & Deliverable
Day 1 (Oct 16) *Next.js 14 Setup*	Scaffold /frontend using Next.js 14 (App Router), Tailwind CSS, Lucide Icons, and Wagmi/Viem. Set up mock state store so UI development proceeds immediately without waiting for backend.	App runs at localhost:3000 with dark slate / emerald theme.
Day 2 (Oct 17) *Biometrics & Dashboard*	Implement **Passkey WebAuthn Login** (1-tap FaceID/TouchID setup; zero seed phrases). Build **Dual-Axis Risk Selector**, **The Tether Radius Gauge** ($ spent vs. daily limit), and the **'Breaking the Tether' FaceID Modal**.	FaceID biometric prompt triggers and signs payload in browser.
Day 3 (Oct 18) *Attack Sandbox & Radar*	Build **The Live Attack Sandbox (/security)** with preset judge exploit buttons (*'Prompt Injection Drain'*, *'High-Slippage Sandwich'*), showing real-time on-chain revert badges from TetherGuardHook. Connect Member 4's PegGuard live news ticker.	Attack Sandbox simulates prompt exploit and displays red revert badge.
Day 4 (Oct 19) *Integration & Polish*	Connect frontend to Member 2's FastAPI endpoints and Member 1's Anvil smart contracts. Rehearse user flow with team for 3-minute video recording.	Butter-smooth live walkthrough ready for screen capture.
Day 5 (Oct 20) *Production Build*	Run production build (npm run build); ensure 0 TypeScript or lint errors. Verify responsiveness across screen sizes.	Production build passing with zero errors.

6. Member 4 Workstream: Financial NLP, Omni-Market Oracles & Simulator

Member 4 builds the PegGuard Layer 4 NLP sentinel, omni-market data feeds, and pre-flight simulation engine:

Day & Timeline	Engineering Tasks & Implementation Scope	Verification Gate & Deliverable
Day 1 (Oct 16) *Omni-Market Feeds*	Build data ingestion pipeline fetching live yields across Web3 DeFi (Lido 3.1%, Aave 4.2%, Curve 6.85%), Chinese Shibor benchmarks (1M/3M/1Y rates @ 2.15%), and Tokenized Gold (PAXG carry @ 1.8%).	GET /api/market/feeds returns normalized multi-market rates.
Day 2 (Oct 17) *PegGuard NLP Pipeline*	Integrate ProsusAI/finbert with Hugging Face Transformers. Build financial news monitor (Bloomberg, Caixin, Reuters, PBOC filings). Implement Named Entity Recognition (NER) matching protocol and issuer entities.	FinBERT accurately scores headlines; filters entity mentions.
Day 3 (Oct 18) *Adverse Risk Veto*	Implement the Asymmetric Risk Veto formula: Circuit Breaker = VETO if P(Negative News) ≥ 60% and Entity Matched. Connect veto trigger to shrink Tether Radius to $0 during breaking exploits.	Simulated bad news successfully triggers emergency trade veto.
Day 4 (Oct 19) *Simulator & Demo Assets*	Build Anvil / Tenderly pre-flight dry-run simulator enforcing slippage ≤ 0.5%. Co-lead demo video recording: capture NLP sentinel veto and cross-market yield comparisons.	Pre-flight simulator integrated; video assets recorded.
Day 5 (Oct 20) *Documentation & Final*	Document FinBERT model weights and NLP methodology in repo; assist team with final pitch deck diagrams and submission email lodgement.	NLP pipeline documented; final submission verified.

7. Zero-Blocker Interface Handoff Protocols

To guarantee that no teammate is ever blocked waiting for code from another member, follow these three explicit handoff interfaces:

Interface Handoff	Between Roles	Interface Format & Location	Zero-Blocker Fallback Protocol
Contract ABIs & Addresses	Member 1 → Members 2, 3, 4	/shared/contracts.json (Anvil contract addresses, compiled JSON ABIs, and EIP-712 domain data).	Members 2, 3, 4 use predefined mock contract ABIs until Member 1 compiles.
Backend REST & NLP APIs	Members 2 & 4 → Member 3	/shared/api_spec.json (FastAPI Swagger OpenAPI schema at /docs for planning & news).	Member 3 uses typed mock JSON objects in Next.js until backend runs.
Live Demo Environment	Member 3 → Whole Team	Running UI at http://localhost:3000 with pre-cached demo wallets & attack presets.	Preloaded demo data allows team to record 3-minute video cleanly in 1–2 takes.

8. Daily 18:00 Team Synchronization Milestones

Date & Day	Member 1 (Contracts)	Member 2 (Cognitive AI)	Member 3 (Frontend)	Member 4 (NLP & Oracles)
Oct 16 (Fri)	TecAccount & Hook written	Mandate parser running	Next.js shell & mocks live	Shibor & DeFi oracles live
Oct 17 (Sat)	Paymaster & Validator written	Dual-Axis risk engine ready	Passkey & Tether Gauge ready	FinBERT news pipeline live
Oct 18 (Sun)	100% Foundry tests green	Proactive strategy engine ready	Attack Sandbox & Radar ready	NLP veto & Anvil dry-run passed
Oct 19 (Mon)	Gas report & tx hashes logged	UserOps submitted to testbed	UI connected to API & testnet	Team records 3-minute video
Oct 20 (Tue)	Code NatSpec verified	Requirements.txt frozen	Production build passed	**Team Submits to Tsinghua**

9. Shared Deliverables & Final Lodgement (Due Oct 20 by 18:00 Beijing Time)

Shared Team Ownership: Pitch deck production, demo video recording, and email submission are shared across the team.
• **Member 1:** Provides contract transaction logs & gas benchmarks.
• **Member 2:** Provides intent parser traces & proactive strategy comparisons.
• **Member 3:** Operates the live running UI at localhost:3000 during video capture.
• **Member 4:** Provides FinBERT sentiment plots, macro yield charts, and coordinates recording & upload.
Official Lodgement Email: Transmitted to wangyr@pbcsf.tsinghua.edu.cn attaching Pitch Deck PDF, Project Spec PDF, GitHub repo URL, 3-minute video demo URL, and team passport scans (≥ 60% non-Chinese).

Tec-C 4-Person Engineering Roadmap & Technical Workstreams. Prepared for the Shenzhen International FinTech Competition (FinTechathon 2026).